THE COHOMOLOGY OF THE CONE ATTACHED TO A FIBRE PRODUCT OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the degree -2 identification and fixes the four local rings in the same display; the proof is a stalk reduction followed by the ring case.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- Commutation of the cotangent complex with stalks and with pullback. Lemmas lemma-stalk-cotangent-complex and lemma-pullback-cotangent-morphism-topoi, used at proof lines 4841-4844. Reduces the statement to a computation at one geometric point.
- The localization results for the cotangent complex. Section section-localization, used at proof lines 4845-4847. Give the identification L_{C/R} = L_{A tensor_R B/R} tensor C.
- The henselization description of the etale local ring of a fibre product. used at proof lines 4845-4845. Identifies C as a henselization of A tensor_R B.
- The ring case of the computation. Lemma lemma-tensor-product (3063-3076, produced as L44133's blocked counterpart; here reported as the ring-level input), used at proof lines 4850-4851. Supplies the vanishing and the degree -2 identification.

Source reference. Original equation 09DK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11868–11868. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08SV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11143–11143. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08T0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11148–11148. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 09DB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11859–11859. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 08QY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11076–11076. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $X \to B$ and $Y \to B$ be morphisms of algebraic spaces over S. The object E in (09DK) satisfies $H^i(E) = 0$ for $i = 0, -1$ and for a geometric point $(\overline{x}, \overline{y}) : \mathrm{Spec}(k) \to X \times_B Y$ we have*

$$H^{-2}(E)_{(\overline{x},\overline{y})} = Tor_1^R(A, B) \otimes_{A \otimes_R B} C$$

where $R = \mathcal{O}_{B,\bar{b}}$, $A = \mathcal{O}_{X,\overline{x}}$, $B = \mathcal{O}_{Y,\overline{y}}$, and $C = \mathcal{O}_{X \times_B Y,(\overline{x},\overline{y})}$.

Proof. The formation of the cotangent complex commutes with taking stalks and pullbacks, see Lemmas 08T0 and 08SV. Note that C is a henselization of $A \otimes_R B$. $L_{C/R} = L_{A \otimes_R B/R} \otimes_{A \otimes_R B} C$ by the results of Section 08QY. Thus the stalk of E at our geometric point is the cone of the map $L_{A/R} \otimes C \to L_{A \otimes_R B/R} \otimes C$. Therefore the results of the lemma follow from the case of rings, i.e., Lemma 09DB. □